

VICTOR HUGO CAMPOS

Full-Stack Developer · Cloud (AWS/GCP) · AI Engineering

linkedin.com/in/vhs4 · github.com/vhs4 · vhs4.dev · contato@vhs4.dev · +55 (11) 94772-0129 · São Paulo, SP — Brazil

PROFESSIONAL SUMMARY

Full-Stack Developer and AI Engineer with production experience across healthtech, edtech, AI and B2B platforms — building scalable, high-performance systems with real business impact. Strong foundation in React, Next.js, Node.js, NestJS, Fastify, Kotlin, AWS and GCP, with deep specialization in AI engineering — multi-agent systems, LangChain, LangGraph and agentic workflows. Track record of leading technical decisions, designing multi-service architectures from scratch, internalizing critical outsourced modules and delivering measurable results — including building a multi-agent conversational AI platform end to end, raising app store ratings by 2+ stars, generating millions of BRL in savings through feature internalization, and winning national and global hackathons. Uses AI-assisted development tools (Claude Code, Antigravity) daily to maximize engineering velocity. Seeking Lead Full-Stack and senior engineering opportunities.

PROFESSIONAL EXPERIENCE

Software & AI Engineer | Forward Deployed Engineer · Barkus (EdTech)

May 2026 – Present

Remote · Contract

- **Overview:** Sole engineer responsible for the end-to-end architecture and development of Lara 2.0 — a conversational financial education platform on WhatsApp powered by a multi-agent AI system, in production with 7,000+ public and private school students and 1.5M+ messages processed. Designed and built the entire technical stack: three independent backend services, a CMS with an AI content-authoring pipeline, a complete teacher portal with admin panel and production infrastructure on GCP. I act as a Forward Deployed Engineer: beyond engineering, I provide technical support to students and the operations team, diagnosing and solving issues directly in production.
- **Multi-agent AI system:** Architected a pipeline of specialized AI agents coordinated through a state machine that drives the entire learning journey. Each agent has dedicated system prompts, context injection and tool access. The system routes student messages to the right agent based on current state, progress and conversation history.
- **Real-time personalization:** Built a system that dynamically adjusts pedagogical tone, difficulty, content framing, engagement strategies and the gamification itself for each student.
- **Three backend services:** ADK (Express + raw SQL/pg) — core AI engine with WhatsApp message routing, agent orchestration and student state management. CMS (Next.js 14 monolith + Server Actions + Prisma 5) — content management with an AI authoring pipeline where Gemini 2.5 Flash generates pedagogical content from topics defined by the pedagogical team, plus full content management, tenant management, teacher lifecycle and gamification configuration. Teacher Portal (Fastify 5 + Prisma 6) — teacher portal with JWT authentication, class management, student enrollment via generated codes, multidimensional rankings (students/teachers/classes) with completion tracking, full audit trail with admin identification, password management and CSV/Excel export.
- **Multi-tenant SaaS:** Designed a comprehensive PostgreSQL schema with full tenant isolation supporting multiple educational institutions in a single deployment. Implemented custom gamification with XP, virtual coins, experience levels, achievement badges and competitive rankings across students, teachers and classes.
- **Admin panel & audit trail:** Built a cross-tenant admin dashboard with a global view — including cross-tenant teacher/student/class rankings with completion filters, full teacher lifecycle management (create, deactivate, reactivate, permanently delete), welcome emails on teacher creation, and a complete audit trail tracking every administrative action with timestamps and admin identification.
- **Support Center:** Created Lara's Support Center, used daily by the (non-technical) support team to solve day-to-day operational issues.
- **WhatsApp integration & infrastructure:** Implemented the complete messaging pipeline integrated directly with Meta's official API (WhatsApp Cloud API) — led the migration that removed Chatwoot from the architecture with zero downtime — with an async queue on Cloud Tasks (retries, DLQ and no student message is ever lost), OpenAI Whisper for audio transcription and LLM observability with Langfuse. Deployed on GCP Cloud Run with Cloud SQL PostgreSQL 16 and Cloud Build CI/CD — a serverless, auto-scaling architecture — plus continuous AI cost optimization in production (prompt caching, observability sampling and elimination of wasteful retries).

Stack: TypeScript · Express · Next.js 14 · Fastify 5 · Prisma · PostgreSQL 16 · Google ADK · OpenAI (GPT-4.1) · Gemini 2.5 Flash · Whisper · GCP Cloud Run · Cloud SQL · Cloud Tasks · Langfuse · Meta WhatsApp Cloud API

Software & AI Engineer • Codrix

Apr 2026 – Jul 2026

São Paulo, SP — Brazil (Hybrid) • Full-time

- **Full-stack AI engineer:** owned the entire stack in the development of two complete AI-based products — from architecture to production deployment.
- **AI-powered B2B Prospecting Platform:** B2B customer prospecting platform with data enrichment and AI lead qualification. Full-stack architecture with Next.js + NestJS + Python; two-stage domain enrichment pipeline (Apify company-domain actor + AI verifier agent); agent built with Agno Framework + GPT-4o-mini + DuckDuckGo for real-time website discovery and verification. ICP system with AI-automated generation and filters by segment, size and state, plus LinkedIn prospecting with an async queue (Supabase + Python worker). Stack: Next.js • NestJS • Python • Supabase • Apify • OpenAI.
- **Multi-Agent AI Video Generation SaaS:** Platform that generates complete commercial videos (9:16 for Reels/Shorts/TikTok) from a text briefing, using a pipeline of 12 specialized agents orchestrated in LangGraph (Director, Casting, Art Direction, Storyboard, Critic, Image Engineer, Video Director, Audio Reconciler, among others) connected in a graph with persistent state and failure recovery. Quality gates with critic agents that score each phase and trigger automatic regeneration loops, plus a self-healing mechanism that detects safety blocks, timeouts and failures and regenerates artifacts without intervention. Videos generated in ~5–10 minutes with narration, synchronized voice-over and character consistency. Stack: TypeScript • LangGraph • Next.js • Fastify • BullMQ • Redis • Postgres + pgvector • Supabase • Docker Swarm • Traefik • ffmpeg • Imagen • Veo • Claude • Gemini.

Software Engineer • Freelance

Mar 2026 – Present

Remote

- **AI-powered Learning Management System:** Designed and delivered a complete LMS (frontend, backend and infrastructure) for a PhD professor at a leading Brazilian university — sole engineer, in partnership with a UX/UI designer; every technology decision and implementation was fully independent.
- **Espherinha — Lead Qualification Agent:** Built and deployed a conversational WhatsApp agent for Esphera Glamping, a luxury hospitality brand with 70K+ Instagram followers and 100+ Google reviews — autonomously qualifying inbound leads. Stack: N8N (queue mode) + WhatsApp Business API (Meta) + Traefik.
- **Esphera Internal AI Concierge:** Architected and deployed a RAG-based internal knowledge assistant trained on the property's amenities, policies, pricing and FAQs — enabling staff to answer guest questions instantly and reducing manual work. Stack: N8N (queue mode) + WhatsApp Business API (Meta) + Traefik.

Full-Stack Developer (Trainee) • SysMap Solutions — Fully Ecosystem (Healthtech)

Apr 2025 – Mar 2026

São Paulo, Brazil (Remote) • Client: Fully Ecosystem — Prudential B2B Platform

- **Platform scale:** Production app with 50K+ active users and 100K+ total users in 3 languages, backed by a BRL 200M investment from Prudential — serving B2B health and wellness solutions.
- **App store ratings:** Raised the iOS rating from 2.8 → 4.3 and Android from 2.5 → 4.9 through continuous technical improvements, performance optimization and UX-focused engineering.
- **Module internalization:** Actively contributed to bringing critical outsourced modules in-house, including the user digital wallet and financial transaction processing — part of a larger initiative that generated millions of BRL in operational savings.
- **Digital wallet:** Built the core backend layer of the user financial module using NestJS and TypeORM — monetary transactions, balance management and payment flows.
- **AWS serverless:** Built Lambda-based automations for operational and analytical report generation, supporting decision-making at scale.
- **Testing standards:** Created and documented a unit and integration testing framework with Jest, adopted company-wide as the official internal guide.
- **Full-stack delivery:** Shipped production backend features using NestJS, Node.js, Kotlin, Docker and AWS (Lambda, S3, ECS); kept close communication with the React Native frontend team to ensure contract and integration alignment — within agile squads with Product, Design and Business.

Tech Lead & Full-Stack Developer • Freelance

Jun 2024 – Apr 2025

Remote

- Led end-to-end delivery of high-impact digital products for clients in the innovation and luxury hospitality sectors, owning architecture, development and technical decisions.

- **Esphera Glamping — Website:** Delivered a premium, responsive, multilingual website (PT/EN/ES) focused on SEO, performance and conversion.
- **Hackathon & Open Innovation Platform:** Led full-stack feature development — new engagement features, scalability improvements and frontend architecture.

Front-End Development Intern • Elastic Code

Jan 2024 – Jun 2024

Remote

- Led the frontend architecture and implementation of a Fleet and Freight Management ERP — driver management, trip cost/weight reports and real-time dashboards.
- Managed global state across multiple unrelated views using Redux, ensuring trip data was simultaneously accessible in driver profiles, trip listings and financial reports without parent-child coupling.
- Stack: React, TypeScript, Tailwind CSS, NextUI, Styled Components, Redux.
- Contributed frontend components to an AI File Analysis Platform integrated with AWS.

Front-End / E-Commerce Developer • Freelance — Shopify

Aug 2023 – Jan 2024

Remote

- Built and customized complete online stores using Shopify Liquid templates — product pages, collections, responsive design and brand identity.
- Delivered conversion-focused storefronts aligned with business goals and consistent across mobile/desktop.

Front-End Development Intern • Campos Developers

Apr 2023 – May 2023

Campos dos Goytacazes, RJ — Brazil

- First professional software development experience — built interfaces with Vue 2, Nuxt, Bootstrap, Buefy, JavaScript, HTML and CSS.
- Implemented CRUD features and REST API integrations; applied Swagger and Redoc for API documentation.

PROJECTS

AI-powered Learning Management System • Full-Stack — Solo architecture and delivery

React • Next.js • NestJS • PostgreSQL • OpenAI GPT-4o • JWT • DDD • Clean Architecture • Docker • Traefik • Cloudflare

- **Solo ownership:** Designed and delivered the entire system alone, with the design crafted by a partner UI/UX Designer — frontend, backend, infrastructure and production deployment — from data model to production environment.
- **Deployment:** Containerized with Docker Compose, with a Traefik reverse proxy (automatic TLS/SSL), served behind Cloudflare for DNS, DDoS protection and CDN.
- **AI quiz generation:** Integrated OpenAI GPT-4o to generate multiple-choice questions from PDFs or teacher-typed content, returning structured, validatable JSON editable before persistence.
- **AI learning-gap analysis:** Built a learning-gap detection pipeline that compiles student quiz performance and sends it to OpenAI — returning personalized diagnostics with weak topics and recommended review areas.
- **Gamification system:** Implemented a points and badges engine — students earn XP by completing quizzes and sections, unlocking achievement badges displayed on their profile.
- **Multi-role access control:** JWT authentication with three permission levels (Admin, Teacher, Student) and fully protected routes on frontend and backend. Password hashing with Argon2id combined with a server-side pepper — never stored in the database.
- **Features:** Teacher dashboard with performance charts, quiz engine with timer (global or per question), bulk student import via CSV, file upload (PDF/PPTX), integrated slide viewer and mobile-first responsive UI.

AI Newsletter Agent • Autonomous agentic pipeline

Python • GPT-4o-mini • Agno Framework • Tavily Search • SMTP • Prompt Engineering

- **Autonomous newsletter generation:** End-to-end AI agent that researches tech news in real time, writes a formatted HTML newsletter and dispatches it via SMTP — zero manual intervention per run.
- **Agentic architecture:** GPT-4o-mini orchestrated via Agno with external tool calling — Tavily Search for RAG-style retrieval, structured prompt engineering for consistent HTML output, and link verification.
- **Human-in-the-loop:** Approval gate with browser preview before sending — the operator keeps full control.

RAG Web Assistant • Intelligent documentation system

Python • LangChain 1.0 • OpenAI GPT-4o • FAISS • Streamlit • Firecrawl

- **Web-to-knowledge pipeline:** Crawls any URL with Firecrawl, converts to Markdown, indexes in FAISS and answers natural-language questions via LangChain 1.0 Runnables — a private ChatGPT for any documentation site.
- Streamlit UI with a clean URL → crawl → index → query flow; modular service/presentation architecture.

EDUCATION

Associate Degree in Systems Analysis and Development • FIAP

2024 – 2025 • Completed

PBL methodology (project-based learning with real projects for partner companies) • Emphasis: Full-Stack Development • Cloud Computing • Artificial Intelligence • Software Engineering

- **Icare Finance (iFood):** Led the development of an ERP built for iFood focused on financial management and data visualization, within the PBL model.
- **Softtek partnership:** Led the development of an anonymous employee sentiment management platform — emotional well-being tracking with privacy, in a project run in partnership with Softtek.
- **Mentorship:** Mentored first-semester students, supporting technical onboarding and adaptation.
- **Course technical foundation:** Software engineering with Scrum, AI and chatbots, Python, Java (DDD), Kotlin (mobile), front-end, relational and non-relational databases (SQL/PL-SQL, NoSQL, data modeling), DevOps and cloud, quality and testing.

AWARDS AND LEADERSHIP

- **Regional Winner & Global Nominee — NASA Space Apps Challenge (2023):** Led the development of an MVP of a social network for scientists in 48h. Hackathon organized in partnership with the Brazilian Space Agency (AEB). Globally recognized among thousands of teams.
- **2nd Place Nationally — CAPES Hackathon (2024):** Led Brazil's 2nd best solution for the CAPES Academic Portal — AI-integrated search and intelligent assistant. 53 teams, 519 participants.
- **Technology & Innovation Leader — Aspire Leaders Program (ongoing):** Program founded by Harvard Business School professors, present in 195+ countries, developing global leaders.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Kotlin, Python, PHP, SQL, HTML5, CSS3

Frontend: React.js, Next.js, React Native, Tailwind CSS, Styled Components, NextUI, Redux, Vue 2, Nuxt

Backend: Node.js, Express, NestJS, Fastify, Kotlin, Java, Django, Flask

Cloud & DevOps: AWS (Lambda, S3, ECS, Cloud Practitioner), GCP (Cloud Run, Cloud SQL, Cloud Build), Docker, Kubernetes, CI/CD

Databases: PostgreSQL, MongoDB, NoSQL, SQL, Data Modeling, PL/SQL, Prisma

AI & Automation: LangChain, LangGraph, OpenAI API, Gemini API, Whisper ASR, N8N, Agentic AI, Multi-Agent Systems, RAG, Prompt Engineering, AI Pipelines, FAISS

AI Tools: Claude Code, Antigravity — daily use for AI-assisted development and engineering velocity

Testing: Jest, Unit Testing, Integration Testing, Supertest, TDD

Methodology: Scrum, Agile, DDD, Clean Architecture, Technical Leadership, Code Review

Tools: Git, GitHub, Figma, Swagger, Postman, Linux, Docker Compose, Chatwoot

CERTIFICATIONS

- AWS Certified Developer – Associate — Amazon Web Services
- AWS Certified Cloud Practitioner — Amazon Web Services
- AWS re/Start — Amazon Web Services
- LangChain: Develop AI Agents — Alura
- AI Upskill Workshop — hands-on AI immersion led by experts trained at Stanford and in Silicon Valley.